Running DRC and LVS

> To follow this doc, you will need to have successfully run the flow to completion.
>
> You will need the files `outputs/apr/DBS/<design_name>.phy.v` and `outputs/apr/DBS/<design_name>.merged.gds`.

DRC and LVS are run in Pegasus, a tool within Cadence Virtuoso.

First, we need to stream our GDS into a new library.

To stream in a GDS:

1. Enter the signoff directory with `cd $wd/signoff`.

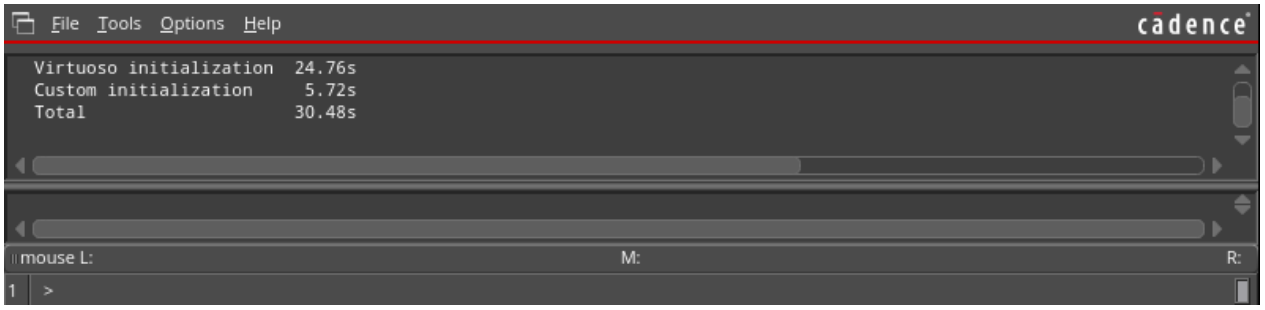
   This directory is prepopulated with a `cds.lib` that contains some libraries we will reference when creating our own.

> Note: If `cd $wd/signoff` didn't work, you're probably not in `SJapr`. Please run `tcsh` and `source envsetup` before continuing.

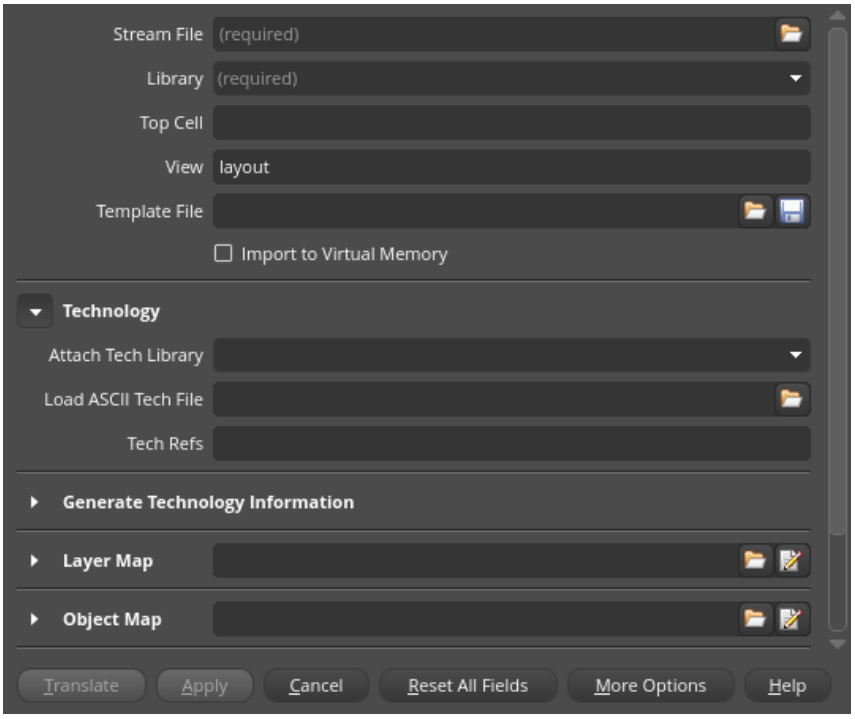
2. In this directory, run `make virtuoso`.

   This will open the Virtuoso GUI. Give it up to a minute to load.

> Note: This opens a GUI. That means we must do this from within FastX.



3. In the GUI, select `File > Import > Stream`. This opens a new window.



4. In the new window, specify the following information:

Field	Value
Stream File	`outputs/apr/DBS/<design_name>.merged.gds` . You will have to click the folder icon, then navigate to the gds file.
Library	A name for your new library (can be anything, just don't select from the dropdown)
Attach Tech Library	`sky130_fd_pr_main` (from the dropdown). This populates some of the other fields automatically.

5. Click `More options` . In the new window, click the folder icon for the `Ref Lib File Name` field and choose `NoCollateral.reflib` .

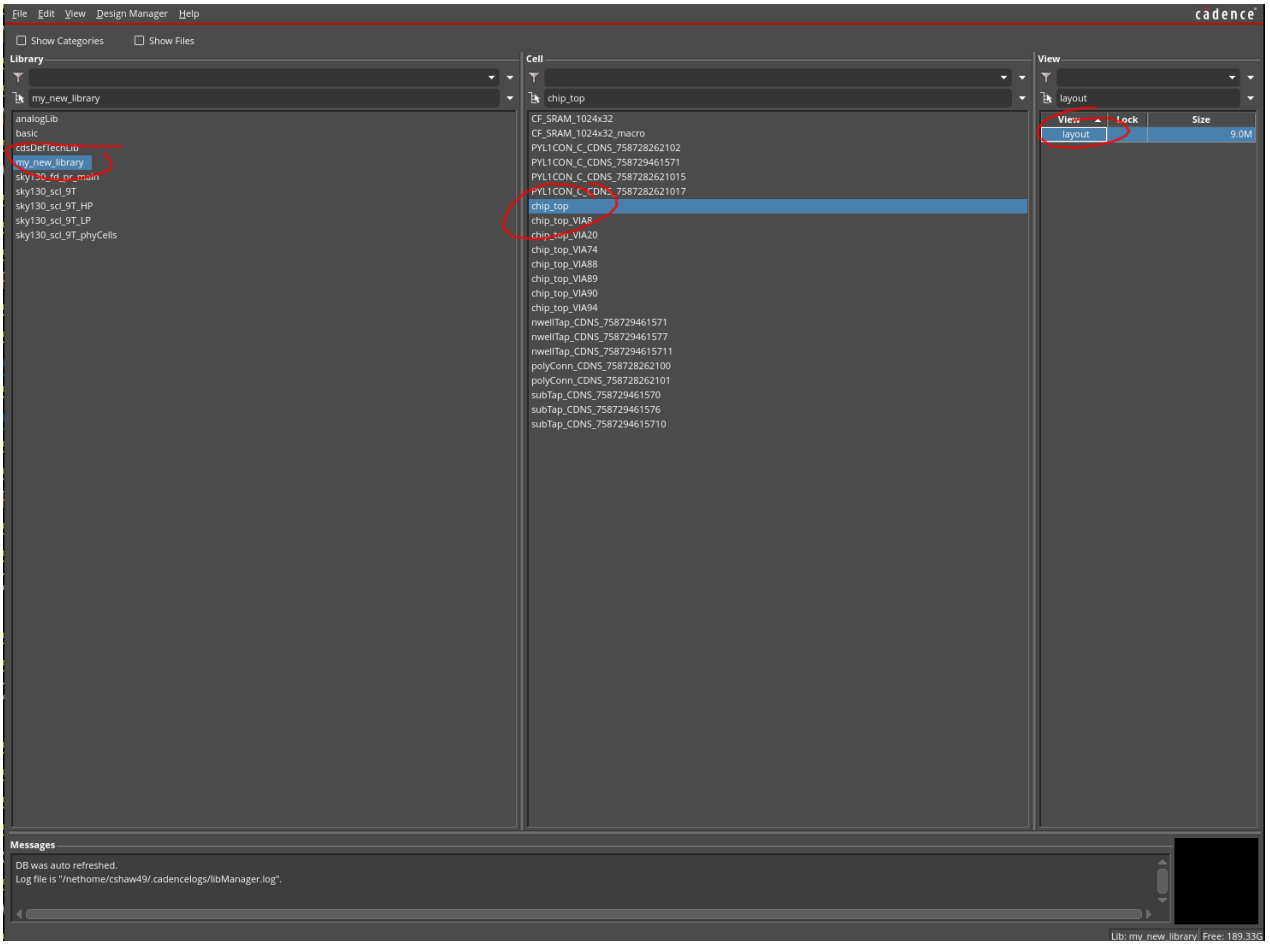
6. Click `Ok` , then click `Apply` . It will load for a moment, then you should get a message that it finished with 0 errors.

   Your library is created.

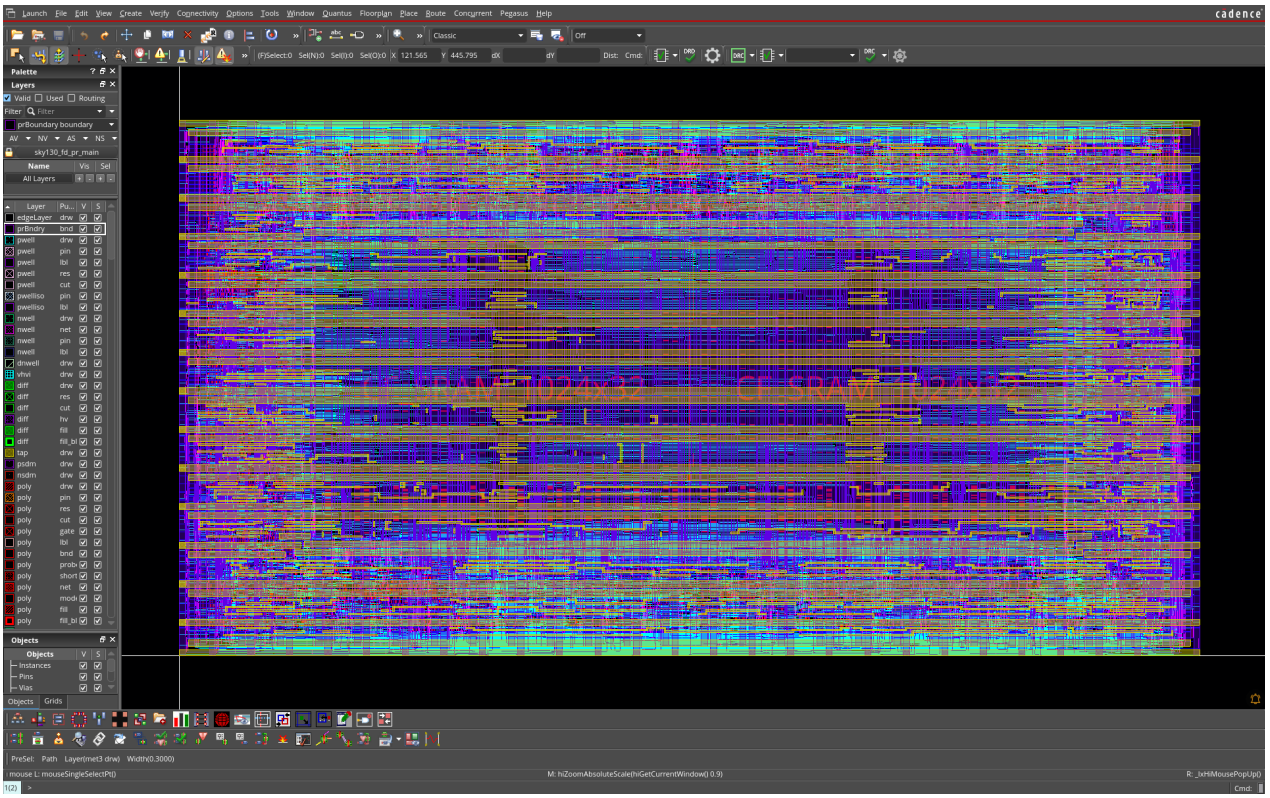
To open your library:

1. In the original Virtuoso menu, choose `Tools > Library Manager` . A new window opens.

2. In the Library Manager window, choose your library, then choose the name of your toplevel cell (probably `chip_top`), then click `layout` .

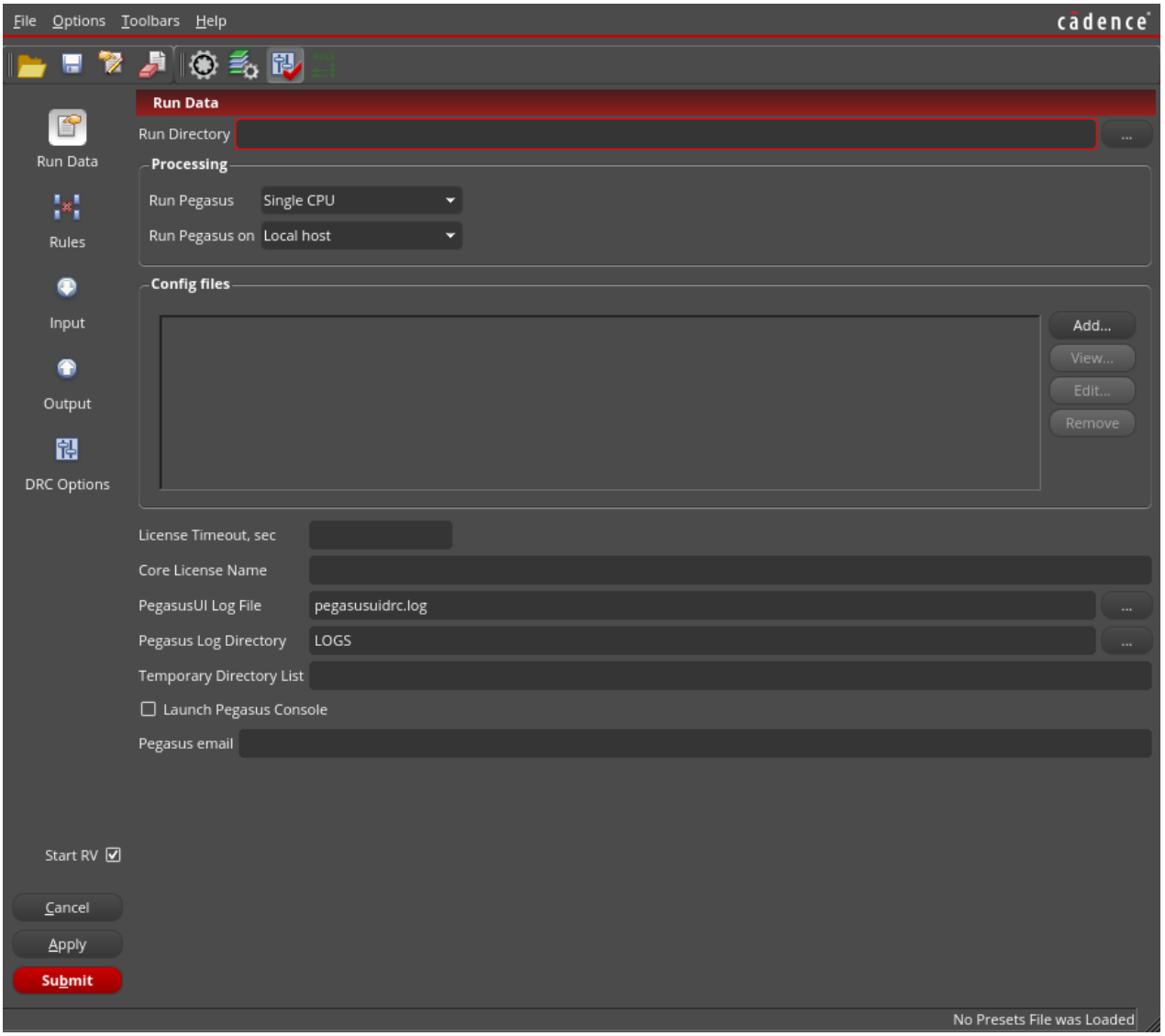


The tool will load for a moment, then your layout will appear.



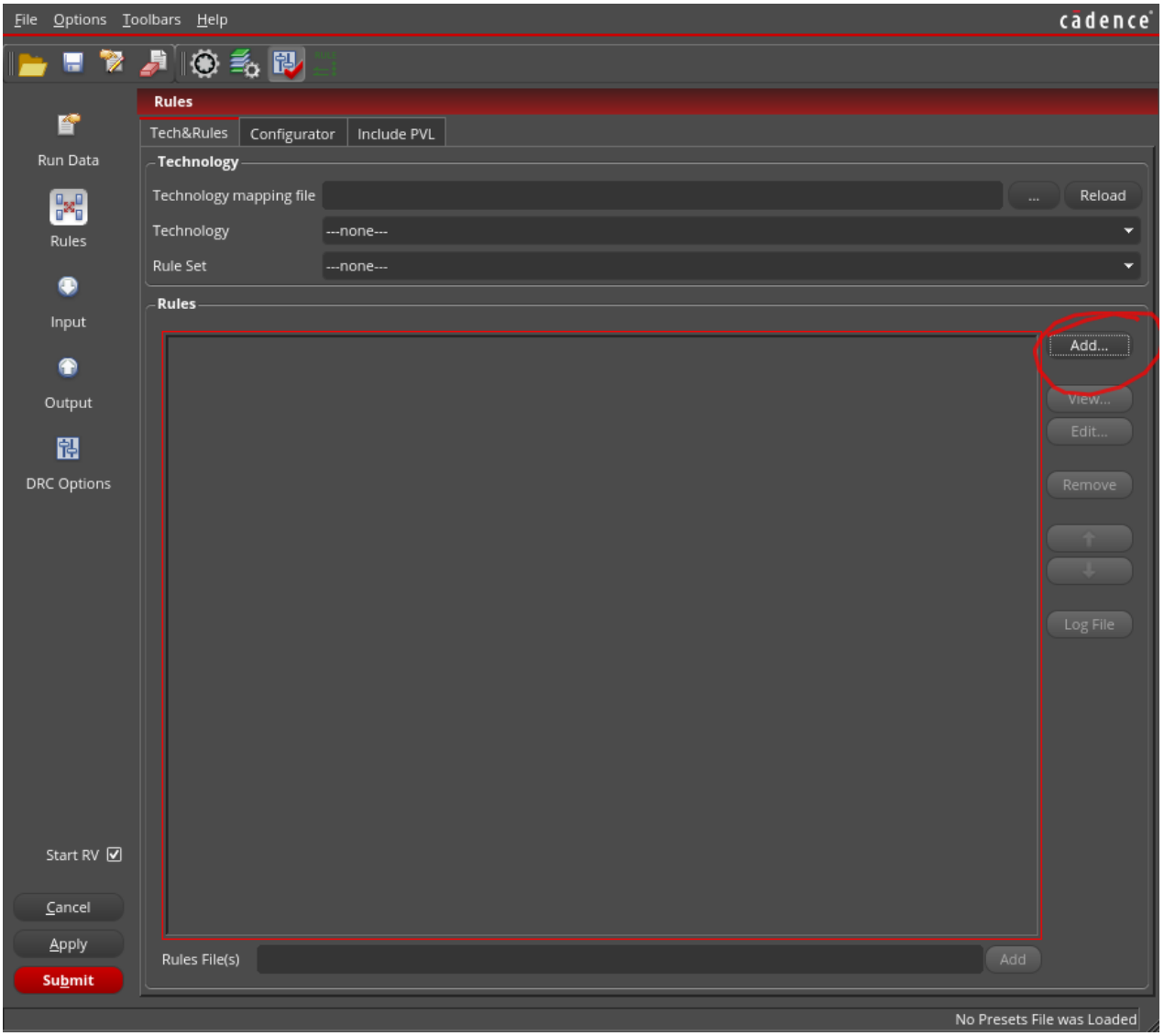
To run DRC on your design:

1. With your design's layout open (as above), click `Pegasus > Run DRC` in the top bar. The Pegasus DRC window opens.



2. In the `Run Data` sidebar, click the three dots next to the `Run Directory` field, and select the `signoff/runs/` directory, then click `Choose` .

3. In the `Rules` sidebar, click the `Add` button.

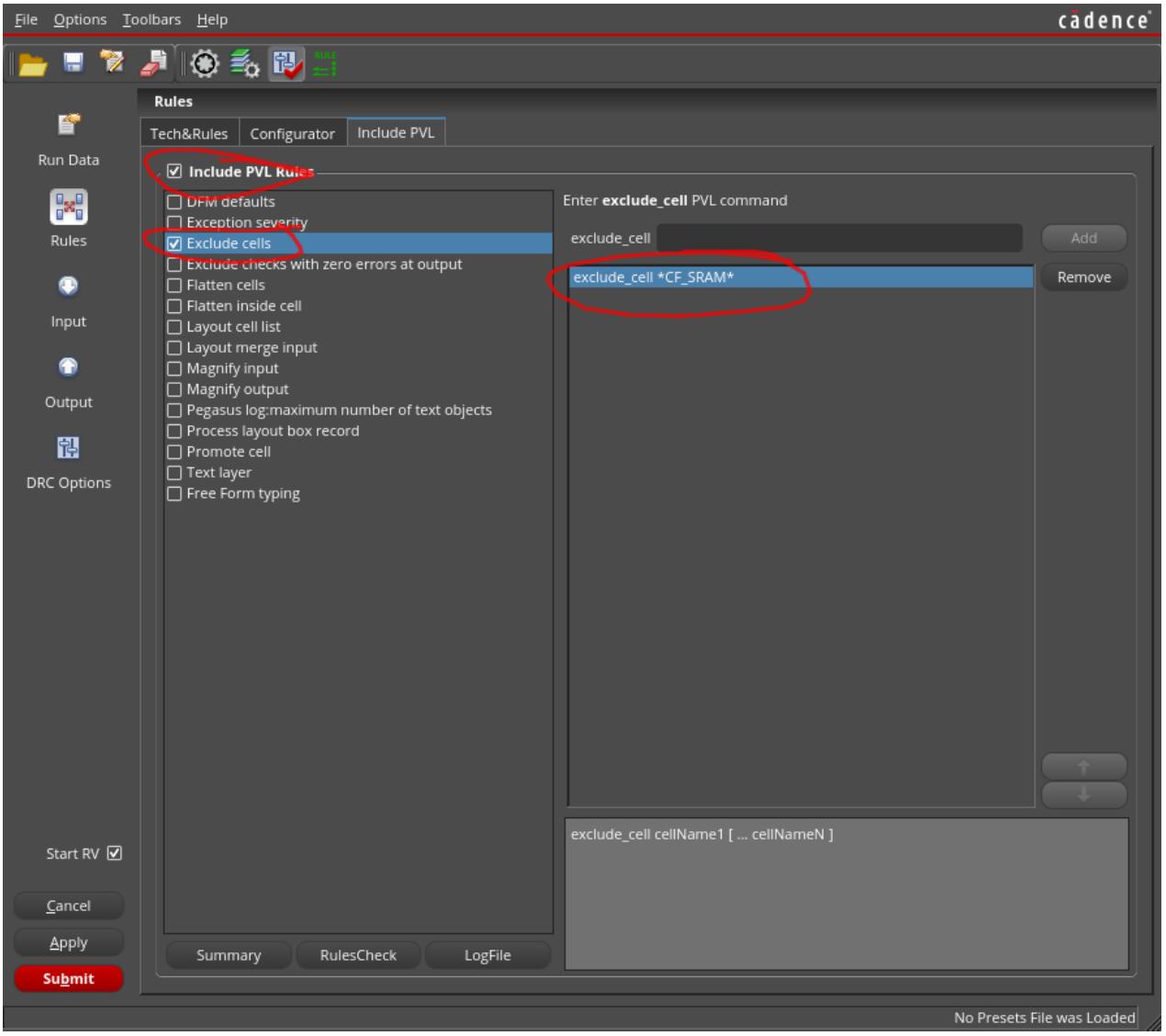


In the popup, navigate to `resources/cadence_sky130/sky130_release_0.1.0/Sky130_DRC/` and choose the file `sky130_rev_0.0_2.12.drc.pvl`, then click `Open`.

In the `Configurator` tab, check the `Use Configurator` box, then click the three dots next to the `Input` field.

In the popup, navigate to `resources/cadence_sky130/sky130_release_0.1.0/Sky130_DRC/` and choose the file `sky130.drc.cfg`, then click `Open`.

In the `Include PVL` tab, check the `Include PVL Rules` checkbox, then check the `Exclude Cells` checkbox. In the `exclude_cell` field on the right, type `*CF_SRAM*`, then click `Add`.

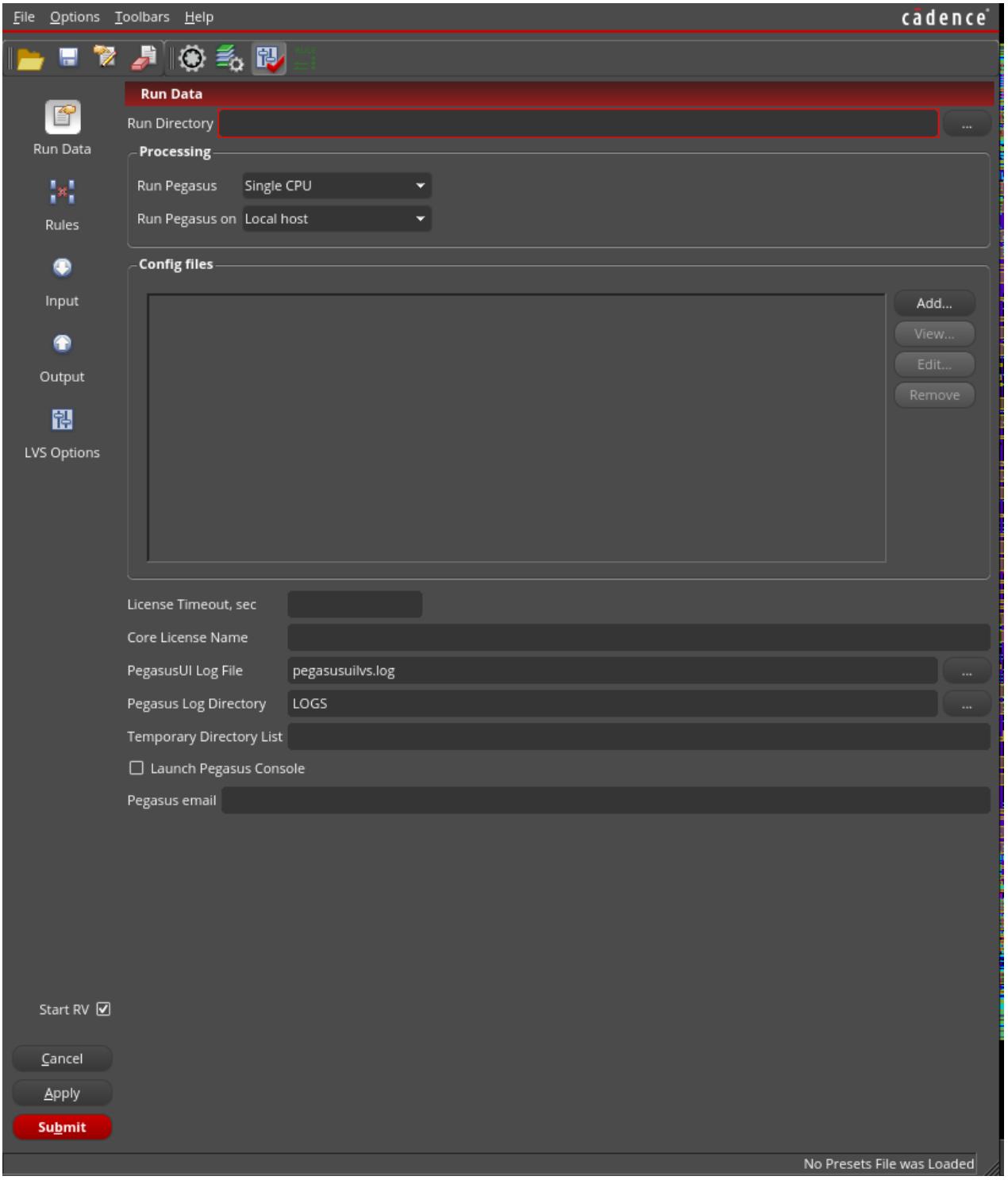


4. Click `Submit` . This will run DRC and generate a report.

> Note: We don't intend for you to fix these DRCs, but it is important that you understand what is causing them.
>
> Try to click the errors in the post-run popup -- the tool will highlight the region the selected error is in in the layout.

To run LVS on your design:

1. With your design's layout open (as above), click `Pegasus > Run DRC` in the top bar. The Pegasus DRC window opens.



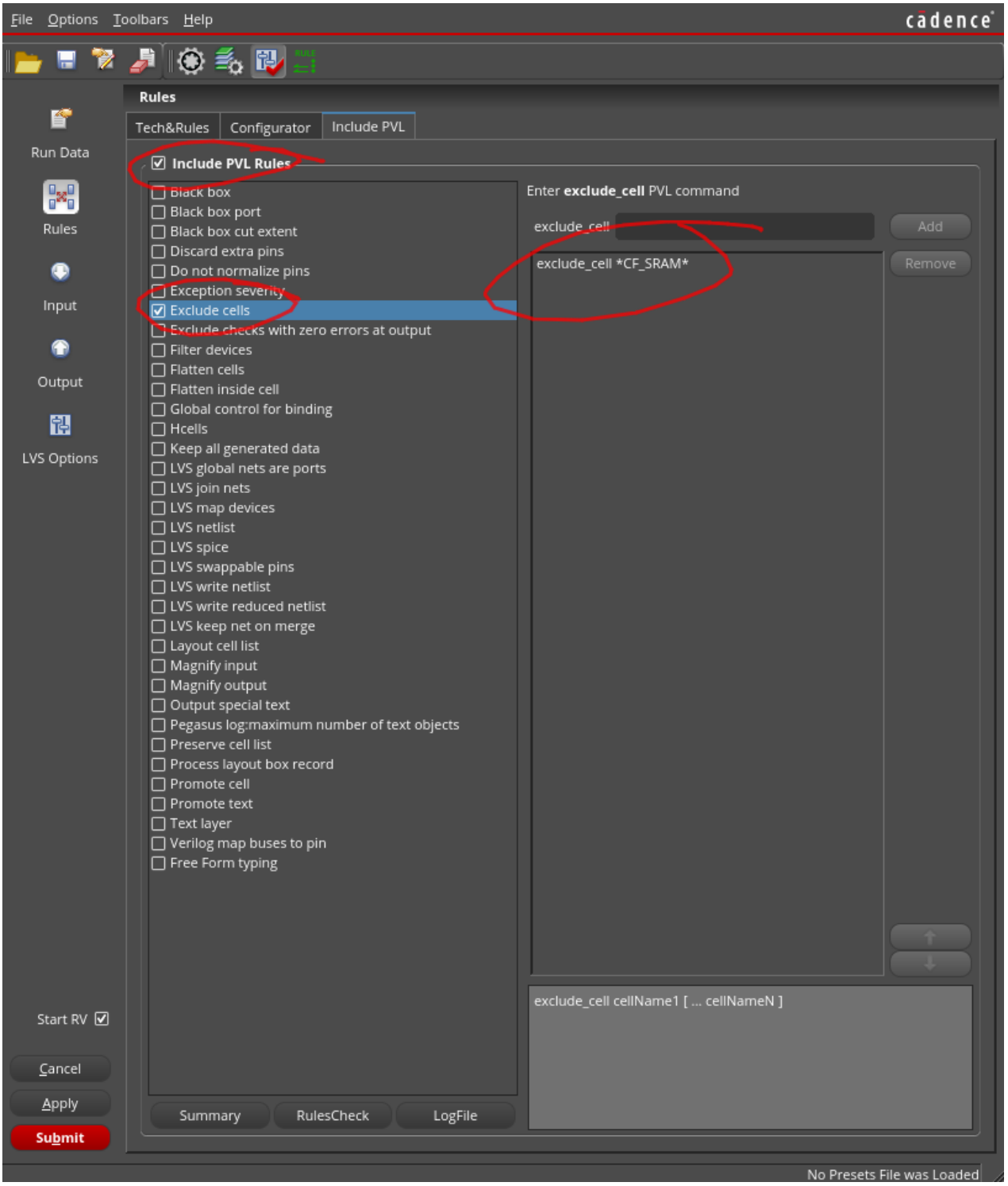
2. In the `Run Data` sidebar, click the three dots next to the `Run Directory` field, and select the `signoff/runs/` directory, then click `Choose`.

> If you just ran DRC, you'll probably get a popup asking if you want to overwrite this directory later. If that happnes, just click `Ok`.

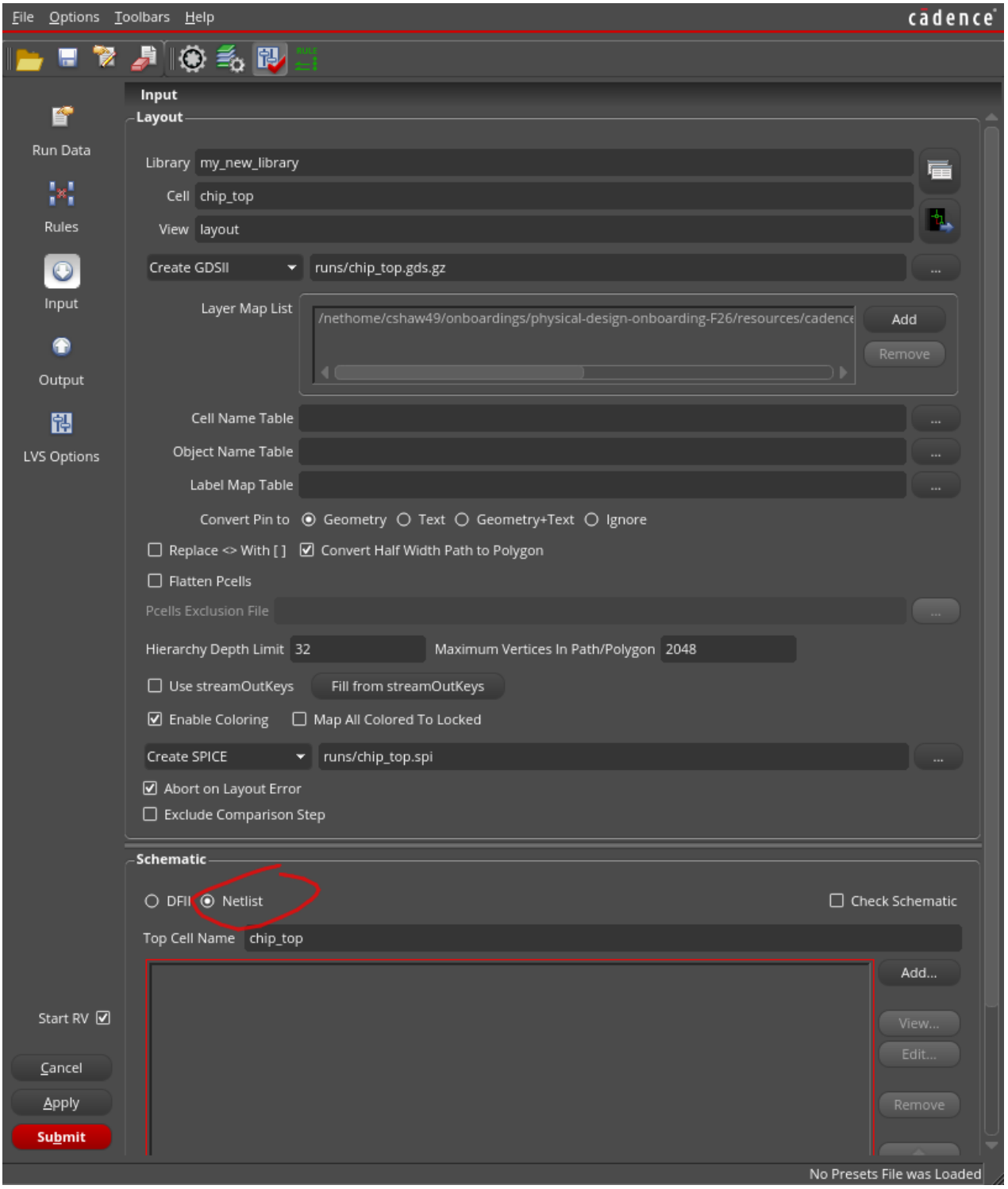
3. In the `Rules` sidebar, click the `Add` button.

   In the popup, navigate to `resources/cadence_sky130/sky130_release_0.1.0/Sky130_LVS/` and choose the file `sky130.lvs.v0.0_1.1.pvl`, then click `Open`.

   In the `Include PVL` tab, check the `Include PVL Rules` checkbox, then the `Exclude cells` checkbox. In the `exclude_cell` field on the right, type `*CF_SRAM*`, then click `Add`.



4. In the `Input` sidebar, click the `Netlist` radio button under `Schematic`.



We have to add a number of files in this section. Add the following:

- `outputs/apr/DBS/<design_name>.phy.v` (LVS is *Layout* vs *Schematic*. The GDS is our Layout, and this file is our Schematic!)
- `signoff/CF_SRAM_1024x32.cdl`
- `resources/cadence_sky130/sky130_scl_9T_0.1.2/sky130_scl_9T/cdl/sky130_scl_9T.cdl`
- `resources/cadence_sky130/sky130_scl_9T_0.1.2/sky130_scl_9T_tech/cdl/sky130_scl_9T_phyCells.cdl`
- `resources/cadence_sky130/sky130_scl_9T_0.1.2/sky130_scl_9T_LP/cdl/sky130_scl_9T_LP.cdl`
- `resources/cadence_sky130/sky130_scl_9T_0.1.2/sky130_scl_9T_HS/cdl/sky130_scl_9T_HS.cdl`

5. Click `Submit`. This will run LVS and generate a report.

> Note: We don't intend for you to fix these LVS errors, but it is important that you understand what is causing them.
>
> Try to read the output and identify the source of the LVS errors.